

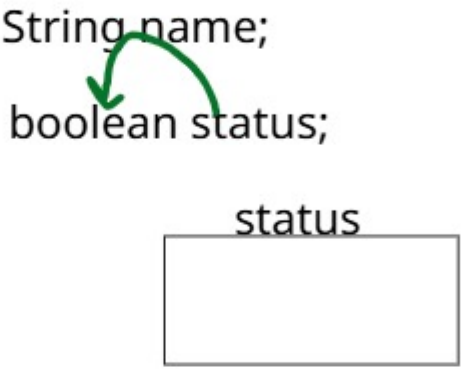
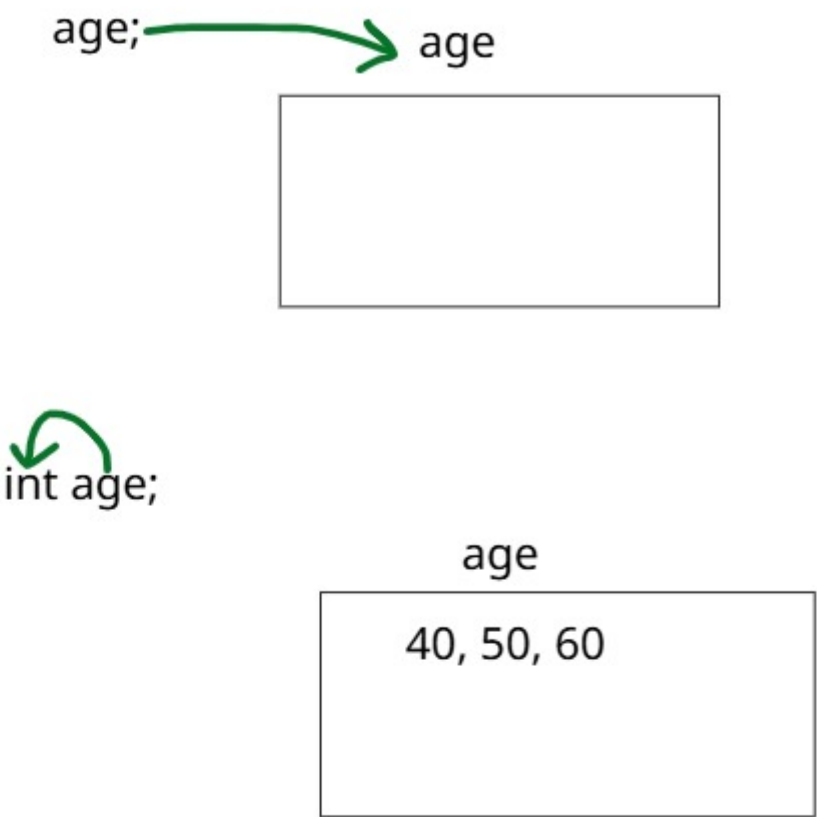
store the age of a person ==> 40

variables are containers / boxes that is used to store some information(name, age, city)

store water ==> container ==> water bottle
store info ==> container ==> variables

store the age of a person ==> 40

create a container(variable)



Whenever we create a container, we should also tell what type of data that container will store



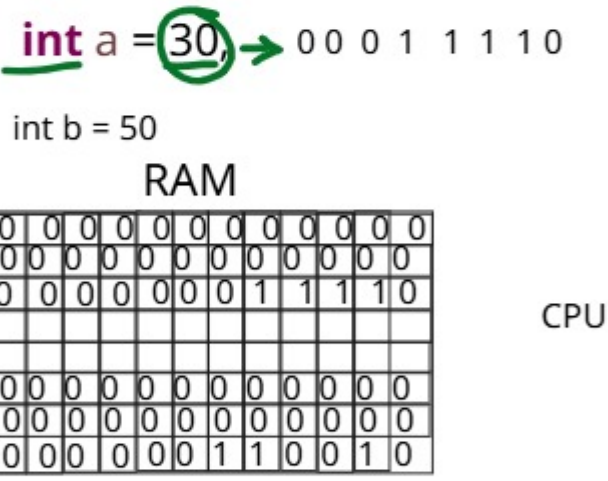
datatype salary;

syntax for variable in java:-

DataType variableName = value

int, byte, short, long

If a variable is of
type byte OR short OR int OR long,
that variable can store only
numbers without decimal point



each box inside RAM ==> 0 or 1

whenever we use int data type,
it gives us 32 boxes

byte ==> give ==> 8 boxes
short ==> give ==> 16 boxes
int ==> give ==> 32 boxes
long ==> give ==> 64 boxes

Data Type	Size	Minimum Value	Maximum Value
byte	1 byte (8 bits)	-128	127
short	2 bytes (16 bits)	-32,768	32,767
int	4 bytes (32 bits)	-2,147,483,648	2,147,483,647
long	8 bytes (64 bits)	-9,223,372,036,854,775,808	9,223,372,036,854,775,807

byte ==> 8 boxes ==> 8 bits ==> 1 byte
short ==> 16 boxes ==> 16 bits ==> 2 bytes
int ==> 32 boxes ==> 32 bits ==> 4 bytes
long ==> 64 boxes ==> 64 bits ==> 8 bytes

float or double ==> can handle numbers with decimal point

float ==> 32 boxes ==> 32 bits ==> 4 bytes
double ==> 64 boxes ==> 64 bits ==> 8 bytes

one alphabet ==> character

char ==> 16 boxes ==> 16 bits ==> 2 bytes